

Shape Optimization

Instructor notes and lab solutions: Sessions 1–11

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English teaching draft

Distribute `lab-sheets.pdf` to students. This document contains solutions, teaching choices and optional proof details. The timed teaching plan and source mapping are in `SESSION_PLAN.md`.

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1 Lab 1: models and distance

Teaching emphasis. Ask for a feasible set before accepting an answer. Separate a minimum value from the point or shape attaining it. The circle comparison is exploratory; the reflection problem receives a complete elementary proof.

Exercise 1.1

(a) Variable $X \in \ell$, objective $|X - Q|$, minimized. (b) Variable a simple closed rectifiable planar curve of length 12, objective enclosed area, maximized. The equilateral triangle has side 4 and area $4\sqrt{3}$, the square has side 3 and area 9, and the circle has radius $6/\pi$ and area $36/\pi$. Only these three candidates have been compared. (c) Variable a rectifiable curve joining A to B whose image intersects ℓ ; objective its length, minimized. Specify whether the endpoints are on the same side of ℓ when applying the reflection construction.

Exercise 1.2

Pythagoras gives $|Q - X|^2 = |Q - H|^2 + |H - X|^2$, with equality only at $X = H$. For $Q = (1, 3)$, the closest point on the axis is $(1, 0)$ and the distance is 3. On the segment $[2, 5] \times \{0\}$, squared distance is $(x - 1)^2 + 9$, smallest at $x = 2$; the distance is $\sqrt{10}$. This example anticipates boundary minima.

Exercise 1.3

For any contact point X , the two portions of any admissible path have lengths at least $|A - X|$ and $|X - B|$. Replacing them by segments cannot increase length. Reflection gives $B' = (6, -4)$ and $|X - B| = |X - B'|$. The segment from $(0, 2)$ to $(6, -4)$ has parameterization $(6t, 2 - 6t)$ and intersects the axis at $t = 1/3$, hence $X = (2, 0)$. The minimum is $|A - B'| = \sqrt{72} = 6\sqrt{2}$. The incoming segment and the reflection of the outgoing segment are collinear, giving equal acute angles with the axis (and with its normal).

For the optional generalization, the contact coordinate is $(ka + hb)/(h + k)$ and the minimum is $\sqrt{(b - a)^2 + (h + k)^2}$. The segment crosses the line because $h, k > 0$.

Exercise 1.4

For $r > 0$, perimeter is $2\pi r$ and has infimum 0, not attained. For area 9π , the radius must be 3 and the perimeter is 6π . Translations of the disk do not change these quantities. A finite numerical search establishes the best value only among the tested radii, subject to numerical error.

2 Lab 2: Torricelli's point

Teaching emphasis. Supply the construction fact explicitly. The task is a geometric proof of global minimality, without differentiating distance or invoking the existence theory from the next session. For a group needing more geometry review, use the supplied 120° point and spend the construction time on perpendicular projections.

Exercise 2.1

The center-to-vertex distance is $s/\sqrt{3}$, so the sum at the center is $s\sqrt{3}$. At a vertex it is $2s$. This gives a better candidate but does not establish a minimum among all X .

Exercise 2.2

Let D be the external equilateral vertex on AB . The minor arc AB of the circumcircle not containing D has central angle 120° , because $\angle ADB = 60^\circ$. A point on that minor arc subtends the complementary 240° arc and therefore sees AB at 120° . The construction uses the corresponding interior arcs for the three sides. The concurrency and interior location are supplied in the exercise, not inferred from a drawing. The argument below only needs an interior point with the three specified angles.

Exercise 2.3

Put the origin at T and write $A = r_A u_A$, $B = r_B u_B$, $C = r_C u_C$, where $r_A, r_B, r_C > 0$ and the u 's are unit vectors separated by 120° . Then $u_A + u_B + u_C = 0$ and $u_A \cdot u_B = -1/2$, cyclically. The three half-planes are

$$u_A \cdot X \leq r_A, \quad u_B \cdot X \leq r_B, \quad u_C \cdot X \leq r_C.$$

Their intersection is a bounded triangle: their outward normals are spaced by 120° and are not contained in a semicircle. Each line is an active side, since its corresponding vertex A , B or C satisfies the other two inequalities strictly. The interior angles of this triangle are all 60° , so it is equilateral. Moreover, all of A, B, C satisfy all three inequalities; for example $u_A \cdot B = -r_B/2 < r_A$. Convexity of the half-planes shows that the whole triangle ABC lies inside this auxiliary triangle K .

For Viviani's identity, let s_K and H_K be the side and altitude of K . If h_1, h_2, h_3 are the distances of an interior point to its side lines, adding the three triangle areas gives

$$\frac{1}{2}s_K(h_1 + h_2 + h_3) = |K| = \frac{1}{2}s_K H_K.$$

Thus $h_1 + h_2 + h_3 = H_K$. The same identity extends to boundary points by the same area decomposition.

Exercise 2.4

Since $A \in d_A$, the distance to the line is no larger than the distance to this particular point: $\text{dist}(X, d_A) \leq |X - A|$. Sum the three inequalities and use Viviani's identity to get

$$F(X) \geq H_K = \text{dist}(T, d_A) + \text{dist}(T, d_B) + \text{dist}(T, d_C) = F(T).$$

The final equality holds because TA, TB, TC are perpendicular to the corresponding lines. Equality for an arbitrary X requires A to be its perpendicular foot on d_A , hence $X \in AT$; similarly $X \in BT$. These distinct lines intersect only at T , proving uniqueness.

An alternative short proof, once dot products are familiar. For any $X \in \mathbb{R}^2$, $|X - A| \geq u_A \cdot (A - X)$ by Cauchy–Schwarz. Summing gives $F(X) \geq r_A + r_B + r_C$ since the unit vectors sum to zero. This even proves minimality over the whole plane. The auxiliary triangle proof presents the same supporting-line idea in geometric form.

Optional coordinate example

Let $t = 1 - 1/\sqrt{3}$. At $T = (t, t)$, $|T - A| = \sqrt{2}t$ and $|T - B| = |T - C| = \sqrt{8/3}$. Computing the dot products of the three unit directions gives $-1/2$ for every pair, so their angles are 120° . The minimum sum is $\sqrt{2} + \sqrt{6}$.

Optional: the angle-at-least- 120° case

Suppose the angle at A is $\alpha \geq 120^\circ$. Write $u = (B - A)/|B - A|$ and $v = (C - A)/|C - A|$; then $\|u + v\| = 2 \cos(\alpha/2) \leq 1$. For $h = X - A$, projection inequalities give

$$|X - B| \geq |A - B| - u \cdot h, \quad |X - C| \geq |A - C| - v \cdot h.$$

Consequently

$$F(X) - F(A) \geq \|h\| - (u + v) \cdot h \geq (1 - \|u + v\|)\|h\| \geq 0.$$

For $\alpha > 120^\circ$ this is strict when $h \neq 0$. When $\alpha = 120^\circ$, equality in both distance projection bounds requires h to lie on both distinct lines AB and AC , so again $h = 0$. This proves uniqueness without differentiating F at a vertex.

Optional return after Session 3: existence of a 120° point

This supplies an analytic existence argument if desired later. On $\mathbb{R}^2$, F is continuous and coercive, since $F(X) \geq 3\|X\| - (\|A\| + \|B\| + \|C\|)$. It is strictly convex when the vertices are noncollinear: equality in the convexity inequality for two distinct points X, Y would force each of A, B, C to lie on the line XY .

A minimizer lies in the closed triangle. Indeed, projection H of an outside point X onto the closed convex triangle satisfies $(X - H) \cdot (V - H) \leq 0$ for each vertex V . Expanding squared distances gives $|X - V|^2 \geq |X - H|^2 + |H - V|^2 > |H - V|^2$. A point in the interior of an edge cannot minimize: a small inward normal motion has zero first derivative for the distances to the two endpoints of the edge, and negative derivative for the distance to the opposite vertex. At a vertex with angle $\alpha < 120^\circ$, the directional derivative in the unit bisector direction is $1 - 2 \cos(\alpha/2) < 0$. Thus, if all angles are below 120° , the unique minimizer is interior.

At an interior minimizer,

$$\nabla F(T) = \frac{T - A}{|T - A|} + \frac{T - B}{|T - B|} + \frac{T - C}{|T - C|} = 0.$$

If unit vectors p, q, r sum to zero, $\|p + q\|^2 = \|r\|^2 = 1$ implies $p \cdot q = -1/2$, and cyclically. Hence all three angles are 120° . This argument is a later complement to the supplied geometric construction, not a prerequisite for Lab 2.

3 Lab 3: existence and elementary minima

Exercise 3.1

(a) Infimum 0, not attained; $x_k = 1/k$. The set is not closed. (b) Infimum 0, not attained; $x_k = -k$. The set is not bounded and the function is not coercive. (c) Minimum 1 at $x = 1$; compactness and continuity apply, or observe that the function decreases on $[0, 1]$. (d) Infimum 0, not attained; $x_k = 1/k$. The set is compact, but continuity and lower semicontinuity fail at 0.

Exercise 3.2

Set $b = A_0/a$, so $p(a) = 2(a + A_0/a)$ for $a > 0$. The function tends to infinity both as $a \downarrow 0$ and as $a \rightarrow \infty$. For a fully explicit compact restriction, compare with $a_0 = \sqrt{A_0}$ and let $M = p(a_0) = 4\sqrt{A_0}$. If $p(a) \leq M$, then $a \leq M/2$ and $a \geq 2A_0/M$. Hence every improving competitor lies in the compact interval $[\sqrt{A_0}/2, 2\sqrt{A_0}]$.

Now $p'(a) = 2(1 - A_0/a^2)$ is negative below $\sqrt{A_0}$ and positive above it. Thus $a = b = \sqrt{A_0}$ uniquely minimizes the side lengths, with perimeter $4\sqrt{A_0}$. Also $(a + b)^2 = (a - b)^2 + 4ab \geq 4A_0$, so $2(a + b) \geq 4\sqrt{A_0}$, with equality precisely at $a = b$. If position and orientation are free, uniqueness is only up to rigid motions.

Exercise 3.3

Squaring is strictly increasing on $[0, \infty)$. For $Q = (1, 2)$, the squared distance is $(t-1)^2 + (2t-2)^2 = 5(t-1)^2$, minimized at $t = 1$ with value 0: the point was already on the line. For $Q = (2, 1)$ the quadratic is $5t^2 - 8t + 5 = 5(t - 4/5)^2 + 9/5$. The closest point is $(4/5, 8/5)$, at distance $3/\sqrt{5}$. Its displacement from Q is $(-6/5, 3/5)$, whose dot product with $(1, 2)$ is zero. Strict convexity proves uniqueness in both cases.

Exercise 3.4

(a) $\nabla f = (2x - 2, 4y + 4)$ gives $(1, -1)$. Since $f = (x - 1)^2 + 2(y + 1)^2 - 3$, the unique global minimum is -3 . (b) $\nabla q = (2x, -2y)$ vanishes only at the origin; values are positive on the punctured x -axis and negative on the punctured y -axis. The origin is a saddle, and q has no global minimum on $\mathbb{R}^2$. (c) The feasible set $\{-1, 1\}$ is not convex; both points minimize x^2 .

Optional: two lines in space

Put $w = A - B$. Differentiating $\|w + tu - sv\|^2$ gives

$$\begin{pmatrix} \|u\|^2 & -u \cdot v \\ -u \cdot v & \|v\|^2 \end{pmatrix} \begin{pmatrix} t \\ s \end{pmatrix} = \begin{pmatrix} -u \cdot w \\ v \cdot w \end{pmatrix}.$$

For independent u, v , the matrix is positive definite: its quadratic form at (a, b) is $\|au - bv\|^2 > 0$ for $(a, b) \neq (0, 0)$. Thus the problem is coercive and strictly convex in the parameters, and the system has exactly one solution. Its displacement is orthogonal to both lines. Parallel lines require a separate discussion and need not have a unique pair of closest points.

4 Lab 4: polygonal isoperimetry

Teaching emphasis. The complete existence argument below is instructor background. State the lemma in the lab, draw the two local gradients at a vertex, and guide the dot-product computation. If the group needs more time, supply the cotangent formula and ask them to explain why it forces equal angles. Leave both extensions for homework.

Exercise 4.1

For $g = ab - A_0$, $\nabla g = (b, a) \neq 0$ in the positive quadrant. The equations are $2 + \lambda b = 0$, $2 + \lambda a = 0$, $ab = A_0$. Thus $a = b = \sqrt{A_0}$ and $\lambda = -2/\sqrt{A_0}$. Lab 3 proves attainment and the global lower bound.

Exercise 4.2

Bisect a central triangle to get a right triangle with opposite side $s/2$, adjacent side r and angle π/n . Therefore $r = s/(2 \tan(\pi/n))$ and $A = nrs/2 = L^2/(4n \tan(\pi/n))$. For $n = 3$ this is $L^2/(12\sqrt{3})$; for $n = 4$ it is $L^2/16$. Since $n \tan(\pi/n) \rightarrow \pi$, the areas approach $L^2/(4\pi)$. This limit of regular-polygon values is consistent with the disk formula; it does not by itself prove the continuous isoperimetric inequality.

Exercise 4.3

The coordinate derivatives of signed area at v_i are

$$\nabla_{v_i} A = \frac{1}{2}(y_{i+1} - y_{i-1}, x_{i-1} - x_{i+1}) = -\frac{1}{2}R(v_{i+1} - v_{i-1}).$$

Differentiating the two adjacent edge lengths gives $\nabla_{v_i} P = e_{i-1} - e_i$. These formulas hold in the open neighborhood of a strictly convex counterclockwise polygon where vertices remain distinct and the polygon remains strictly convex.

Differentiation of $P(tv) = tP(v)$ and $A(tv) = t^2A(v)$ at $t = 1$ yields the two identities in the sheet. In particular, $\nabla P \neq 0$ because $P = L > 0$. Apply the multiplier theorem to minimizing $-A$ under $P = L$: $-\nabla A + \lambda \nabla P = 0$. Taking a scalar product with the full vertex vector gives $2A = \lambda P$, hence $\lambda > 0$.

Exercise 4.4

The vector $\nabla_{v_i} A$ is perpendicular to the diagonal $v_{i+1} - v_{i-1}$. The scalar product of $\nabla_{v_i} P$ with that diagonal is

$$\begin{aligned} (e_{i-1} - e_i) \cdot (\ell_{i-1}e_{i-1} + \ell_i e_i) \\ = (\ell_{i-1} - \ell_i)(1 - \cos \theta_i). \end{aligned}$$

As $\lambda > 0$ and $0 < \theta_i < \pi$, consecutive lengths must agree, so all equal s . Taking norms of the multiplier equation gives

$$s \cos(\theta_i/2) = 2\lambda \sin(\theta_i/2), \quad \lambda = \frac{s}{2} \cot(\theta_i/2).$$

Cotangent is strictly decreasing on $(0, \pi/2)$, so all θ_i are equal. The exterior angles of a convex polygon sum to 2π , giving $\theta_i = 2\pi/n$. Equal sides and equal successive turns describe a regular convex polygon.

The existence lemma supplies a strictly convex maximizer; the multiplier calculation forces it to be regular. Therefore the maximum area for perimeter L is exactly $L^2/(4n \tan(\pi/n))$. Rescaling proves the equivalent fixed-area minimum-perimeter statement. Equality fixes the regular shape and scale, but permits translations and rotations (and reflection if the vertex orientation is not prescribed).

Instructor background: the existence and nondegeneracy lemma

1. A compact parameter space. Consider ordered closed chains with n vertices, allowing repetitions, and perimeter at most L . Fix $v_1 = 0$ to remove translations. Every vertex has distance at most L from 0, and the condition $\sum_i \|v_{i+1} - v_i\| \leq L$ is closed. Thus this set of tuples is compact in $\mathbb{R}^{2(n-1)}$.

Maximize the area of the convex hull of the vertices on this compact set. That area depends continuously on the tuple: convergence of finitely many vertices gives convergence of their convex hulls in Hausdorff distance, and area is continuous for bounded planar convex sets. An elementary justification uses outer parallel sets: if the Hausdorff distance is at most ε , then each hull is contained in the other plus a disk of radius ε . For a convex polygon, the area of this parallel set is its area plus ε times its perimeter plus $\pi\varepsilon^2$, as follows by adjoining strips and circular sectors. Perimeters of the bounded hulls are uniformly bounded, so the area differences tend to zero. Degenerate hulls follow by the segment/disk limit of the same estimate.

2. Passing to a convex polygon loses no perimeter budget. The perimeter of the convex hull of any closed chain is no larger than the chain length. One justification projects the chain onto a line of unit direction u : the total variation of this closed projected chain is at least twice the hull width $w(u)$. Integrating over angles $0 \leq \theta \leq \pi$, an edge of length ℓ contributes $\int_0^\pi \ell |\cos(\theta - \alpha)| d\theta = 2\ell$. For the convex hull boundary, the projected variation is exactly twice its width. Comparing the two integrals gives the claimed perimeter inequality. This is also the familiar “a taut loop encloses a convex hull” observation, with a precise polygonal justification.

The maximum hull area is positive because an equilateral triangle of perimeter L is an allowed competitor. Order the extreme hull vertices around its boundary. This gives a convex polygon with at most n vertices and perimeter at most L . Repeated vertices pad it to an admissible n -tuple if necessary; translation restores $v_1 = 0$. If its perimeter were less than L , scaling up would increase its positive area, contradicting maximality. Hence the perimeter equals L .

3. A maximizer uses all n vertices. Suppose it has $k < n$ genuine extreme vertices. Choose a side of length ℓ and replace it by two segments meeting at a point a distance $h > 0$ outward from its midpoint. For sufficiently small h , the new polygon is still convex, has $k + 1 \leq n$ vertices and has

$$A_h = A + \frac{\ell h}{2}, \quad P_h = L + 2\sqrt{(\ell/2)^2 + h^2} - \ell = L + O(h^2).$$

Rescale by L/P_h to restore perimeter L . Its area becomes

$$\left(\frac{L}{P_h}\right)^2 A_h = A + \frac{\ell h}{2} + O(h^2) > A$$

for sufficiently small h , a contradiction. Thus it has exactly n extreme vertices. All edges are nonzero and no vertex has interior angle π . It is strictly convex in the polygonal sense needed for the local multiplier computation.

Scope. This proves the theorem for convex polygons, which is the lab’s stated class. For a simple nonconvex n -gon, its convex hull has at least as much area, at most its perimeter and at most n vertices. The same convex maximization result therefore extends the inequality to simple polygons. A genuinely nonconvex polygon has strictly less area than its hull, so equality still requires regularity.

Optional extension A: two constraints

The gradients $(1, 1, 1)$ and $(1, -1, 0)$ are independent. With $\mathcal{L} = x^2 + y^2 + z^2 + \lambda(x + y + z - 1) + \mu(x - y - 1)$, the equations yield

$$(x, y, z) = (5/6, -1/6, 1/3), \quad \lambda = -2/3, \quad \mu = -1.$$

The minimum is $5/6$. For a direct check parameterize the feasible line by $(x, y, z) = (t + 1, t, -2t)$. Then $f = 6t^2 + 2t + 1 = 6(t + 1/6)^2 + 5/6$. Closedness and coercivity give existence; strict convexity gives uniqueness.

Optional extension B: Fermat revisited

For X away from the three vertices, $\nabla F(X) = \sum_{V \in \{A, B, C\}} (X - V)/|X - V|$. The unit-vector calculation from the end of the Lab 2 solutions proves the 120° condition. Convexity turns a differentiable stationary point into a global minimizer. At a vertex the corresponding distance term is not differentiable, so the displayed gradient formula cannot be used there; use the projection inequality from Lab 2 instead.

5 Lab 5: scaling, polygons and Dido

Teaching choice. Polygonal approximation is a feasible core argument here because the finite-dimensional polygon theorem has already been developed in Lab 4. State the area-approximation lemma, then prove the three-line limit argument at the board. The cap estimate below is optional background, or a short extension for students comfortable with geometric area bounds. No compactness theorem for a space of general shapes is required for this route.

Exercise 5.1

Since $A = ns^2/(4 \tan(\pi/n))$ and $P = ns$, elimination gives the two formulas on the sheet. For $n = 3, 4, 6, 12$, the perimeters at unit area are approximately 4.55901, 4.00000, 3.72242, 3.58630; the areas at unit perimeter are 0.04811, 0.06250, 0.07217, 0.07775. These computations suggest a disk, but they compare only regular polygons. There is still an entire class of competitors to control.

Exercise 5.2

Differentiate to get $h' = 1 - \cos(2t) = 2 \sin^2 t$. Since $h(0) = 0$, $h(t) > 0$ for $t > 0$ in the stated range. Thus

$$(\tan t/t)' = h(t)/(t^2 \cos^2 t) > 0.$$

Substitute $t = \pi/n$: the direction of monotonicity reverses as n increases. The limit $\tan t/t \rightarrow 1$ gives $n \tan(\pi/n) \rightarrow \pi$. The perimeter at fixed area strictly decreases to $2\sqrt{\pi A_0}$; the area at fixed perimeter strictly increases to $L^2/(4\pi)$.

Exercise 5.3

(a) Dilating H by $10/9$ gives area $6(10/9)^2 = 200/27 > 6$ at perimeter 10, contradicting the premise. (b) Dilate K_* by 2 to obtain area 24 and perimeter 20. If any area-24 shape had perimeter $p < 20$, shrinking it by $1/2$ would produce an area-6 shape with perimeter $p/2 < 10$, contrary to (a)'s argument for arbitrary competitors. (c) A dilation can move a shape outside the fixed square, so it is not necessarily an admissible comparison.

Exercise 5.4

$$A(Q_m) \leq \frac{P(Q_m)^2}{4m \tan(\pi/m)} \leq \frac{L^2}{4m \tan(\pi/m)} \rightarrow \frac{L^2}{4\pi}.$$

Area convergence yields the bound for K . The hull of a nonconvex Jordan region has no smaller area and no larger perimeter, so the same bound applies to the original region. A disk with radius $L/(2\pi)$ attains it. A strict inequality at every polygonal stage does not give strictness in the limit, so uniqueness of equality shapes remains to be proved. This is the role of Session 6.

Exercise 5.5

The reflected region has perimeter 24 and area $2A$, hence $2A \leq 24^2/(4\pi)$, or $A \leq 72/\pi$. The half-disk radius is $12/\pi$ and the used shore segment has length $24/\pi$. A closed fence of length 12 away from the shore encloses at most $36/\pi$, exactly half as much. The free shore contributes the diameter without consuming fence; it is not charged in the reflected outer boundary either.

Approximation lemma: an explicit elementary area estimate

Parameterize the boundary of a convex region K by arc length and split it into m arcs, each of length $\delta = L/m$. Every point of one arc is at distance at most δ from its first endpoint. The disk of radius δ at that endpoint is convex, so it contains the convex hull of the arc, and hence the cap between that arc and its chord. Each cap has area at most $\pi\delta^2$.

The inscribed polygon and these m caps cover K (with disjoint interiors apart from degenerate caps). Consequently

$$0 \leq A(K) - A(Q_m) \leq m\pi\delta^2 = \frac{\pi L^2}{m} \rightarrow 0.$$

The sum of the chord lengths is at most the sum of the arc lengths L . Removing collinear vertices does not change the hull or its perimeter. For large m , the area bound ensures $A(Q_m) > 0$, so the polygonal theorem applies. This proves exactly the approximation properties used in the course, with a deliberately coarse but sufficient error bound.

Convex hull perimeter: why straightening detours is valid

The projection argument in the Lab 4 solutions extends to a rectifiable closed curve. Parameterize it by arc length as $\gamma(s)$, $0 \leq s \leq L$. Its total variation in projection onto the unit direction $u_\theta = (\cos \theta, \sin \theta)$ is

$$V(\theta) = \int_0^L |\gamma'(s) \cdot u_\theta| \, ds.$$

This is at least twice the projection width $w(\theta)$: a closed projected path must travel from its minimum to maximum and back. Integrating over $0 \leq \theta \leq \pi$ gives $\int_0^\pi V(\theta) \, d\theta = 2L$, because a unit vector has $\int_0^\pi |v \cdot u_\theta| \, d\theta = 2$. For the convex hull boundary the projected variation is exactly twice its width, so its perimeter is $\int_0^\pi w(\theta) \, d\theta \leq L$. This also follows first for polygons and then by convex boundary approximation. Rectifiable curves have the arc-length derivative almost everywhere; that integration detail can be left outside the core lesson.

If a compact Jordan region is nonconvex, its convex hull contains an interior point outside the region. Otherwise closedness would force it to contain the whole hull. Because the region is closed, a small disk around a suitable such point is still outside it and inside the hull. The hull therefore has strictly larger area.

Dido reference correction

In the local Treibergs PDF, Section 2, p. 3, the displayed prose value for the maximum area along a straight shore is missing a factor $1/2$. With his stated convention that L is the length of the free arc, the correct value is $L^2/(2\pi)$: either double the area by reflection, or compute $\frac{1}{2}\pi r^2$ with $\pi r = L$. The course and exercises use this convention consistently.

6 Lab 6: bisectors, hinges and logical scope

Teaching choice. Use the half-region hinge as the main geometric argument. Present the quadrilateral method as a second perspective, with the area formula supplied and its derivation left as an extension. Both methods need the local admissibility discussion. An entire flex to the area-maximizing triangle or quadrilateral is not required.

Exercise 6.1

(a) If one free arc had length $\ell < L/2$, doubling its half-region would give area A and perimeter 2ℓ . Dilating by $L/(2\ell)$ would give area $A[L/(2\ell)]^2 > A$ at perimeter L . Thus both arcs have length $L/2$. (b) If equal-length arcs enclosed unequal half-areas, doubling the larger half would increase area while keeping perimeter L . (c) Equal half-areas and equal arc lengths are scalar equalities. They do not identify the entire boundary curves. Reflection constructs a symmetric competitor with the same objective value; it does not prove that the initial shape was already symmetric.

For the existence of an area bisector, put the bounded region inside a rectangle of transverse width D . Moving a parallel cutting line by ε changes the cut-off area by at most $D|\varepsilon|$. The cut-off area is continuous, so the intermediate value theorem applies.

Exercise 6.2

The variable area is $q(\theta) = ab \sin \theta/2$ and $q'(\theta) = ab \cos \theta/2$. For $a = 3, b = 4, \theta = \pi/3$, the triangle area is $3\sqrt{3}$; at a right angle it is 6. The reflected full-region increase would be $12 - 6\sqrt{3} > 0$ if this entire motion were performed admissibly. The proof only uses the positive increase from a sufficiently small motion toward the right angle.

At a maximum every interior boundary point C must give a right angle. Thales' converse puts all such points on the circle with diameter AB . The continuous arc connects A to B in one half-plane and therefore is the entire corresponding semicircle. Both original halves have the same chord, so they form one disk. Check that the moved arcs remain simple, stay above the base, and do not overlap each other or the central triangle except on their intended shared edges. The final competitor is allowed to be nonconvex.

Why a sufficiently small hinge motion is admissible

We apply the constructions only after proving that a hypothetical optimizer is strictly convex. Distinct selected boundary points form a nondegenerate triangle or strictly convex quadrilateral. The caps are the parts of the convex region cut off outside the selected chords.

Away from small neighborhoods of the finitely many hinges, distinct pieces that must remain disjoint are compact and separated by a positive distance. Sufficiently small rigid motions preserve that separation. Near a hinge, a convex body with interior has distinct outgoing one-sided tangent rays, enclosing a positive interior angle. The adjacent short boundary arcs lie in disjoint narrow cones around those rays. Small rotations preserve this local separation. This argument also applies at a smooth point, where the two outgoing rays are opposite.

For the half-region construction, the base chord crosses the interior of the body. The boundary arc at each endpoint therefore meets the base with a positive one-sided angle. After a small motion the arc still lies above the new base near its endpoints; away from them compactness gives a positive distance from the base. Place the moving base on the original cutting line by an

overall rigid motion. Its endpoints may slide, as the comparison does not prescribe the chord length.

Thus a small flex gives another Jordan region with the same free arc length. Rigid cap motions preserve their areas and chord lengths, so the changing enclosed area is exactly that of the central triangle or quadrilateral. A larger flex may create intersections and is unnecessary. For a first presentation, draw this local construction on a smooth strictly convex example; the one-sided tangent explanation covers corners without assuming smoothness of the optimizer.

Exercise 6.3

The correction term in the general formula is nonnegative. Since $a, b, c, d > 0$, equality requires $\cos((\alpha + \gamma)/2) = 0$. For a convex quadrilateral $0 < \alpha, \gamma < \pi$, so this is exactly $\alpha + \gamma = \pi$, the cyclic condition. For a side-2 rhombus with angles $60^\circ, 120^\circ$, $s = 4$ and $Q = 4 \sin 60^\circ = 2\sqrt{3}$, so $Q^2 = 12$. The opposite-angle correction is $16 \cos^2 60^\circ = 4$, giving $Q^2 = 16 - 4 = 12$. For the square it vanishes and $Q = 4$.

The quadrilateral theorem used in the hinge proof must be established independently, for example by the cosine-law calculation below. Treibergs also gives a motivating argument in the other direction, starting from the planar inequality and attaching circular caps. That argument is an application of the isoperimetric inequality, not an independent ingredient for its proof.

Exercise 6.4

Differentiating the cosine laws gives $\alpha'(t) = t/(ab \sin \alpha)$ and $\gamma'(t) = t/(cd \sin \gamma)$. Hence

$$Q'(t) = \frac{t}{2}(\cot \alpha + \cot \gamma) = \frac{t \sin(\alpha + \gamma)}{2 \sin \alpha \sin \gamma}.$$

The denominator is positive, and $\alpha + \gamma \in (0, 2\pi)$. The derivative vanishes exactly when the quadrilateral is cyclic. At any other configuration, choose the sign of a small change of t to increase Q . Nondegenerate convex quadrilaterals form an open set of these local configurations, so small changes are possible.

Three boundary points of a strictly convex region determine a circle. If the region is not a disk, choose a fourth point off that circle and order the four cyclically along the boundary. Their quadrilateral is strictly convex and noncyclic. Keep its sides and attached caps rigid: outer arc lengths and cap areas are fixed, so increasing the central quadrilateral area increases the full enclosed area at fixed perimeter.

Exercise 6.5

For $n \geq 2$, $J(n+1) - J(n) = 1/[n(n+1)] > 0$. Nevertheless $\sup_{n \geq 1} J(n) = 1$ is not attained, and $J(1) = 0$ is the smallest value. The only unchanged point of a chosen improvement rule need not be a global maximizer. Showing that every non-disk admits an improvement excludes non-disks as maximizers; it does not establish that a maximizer exists or that iteration converges to a disk.

Optional: derive the quadrilateral area formula

Let $t = |AC|$, with opposite angles α at B and γ at D . Write $U = ab$, $V = cd$, $\Delta = a^2 + b^2 - c^2 - d^2$. Area addition and the cosine laws give

$$2Q = U \sin \alpha + V \sin \gamma, \quad 2U \cos \alpha - 2V \cos \gamma = \Delta.$$

Square and regroup to obtain

$$\begin{aligned} 16Q^2 &= 4U^2 + 4V^2 - \Delta^2 - 8UV \cos(\alpha + \gamma) \\ &= 4(U + V)^2 - \Delta^2 - 16UV \cos^2\left(\frac{\alpha + \gamma}{2}\right). \end{aligned}$$

The polynomial part factors as

$$\begin{aligned} 4(U + V)^2 - \Delta^2 &= [(a + b)^2 - (c - d)^2][(c + d)^2 - (a - b)^2] \\ &= (a + b + c - d)(a + b - c + d) \\ &\quad \cdot (a - b + c + d)(-a + b + c + d) \\ &= 16(s - a)(s - b)(s - c)(s - d). \end{aligned}$$

Divide by 16 to obtain the general formula, with Brahmagupta's formula as its cyclic equality case.

Optional: existence of a cyclic quadrilateral with given sides

Suppose the positive side lengths satisfy that each is less than the sum of the other three. The two triangles with common diagonal t are nondegenerate for

$$\max\{|a - b|, |c - d|\} < t < \min\{a + b, c + d\}.$$

This interval is nonempty by the side inequalities. In it, the opposite angles $\alpha(t), \gamma(t)$ from the cosine law increase continuously with t . At the lower limiting endpoint their sum is less than π ; at the upper endpoint it is greater than π . Thus a unique value of t gives $\alpha + \gamma = \pi$. Glue the two triangles on opposite sides of the diagonal. Their opposite angles are supplementary, so the four vertices lie on one circle in boundary order. This is the required convex cyclic quadrilateral, and the general formula proves its maximal area for these sides.

7 Lab 7: distances and selection

Teaching choice. Prove the distance-function identity and the selection theorem in the course. State Arzelà–Ascoli in its compact metric-space form; a proof is optional background below. Keep ordinary Hausdorff convergence of compact sets distinct from Henrot–Pierre’s convergence of open sets through their compact complements.

Exercise 7.1

- (a) A closed disk and that disk with an isolated external point are distinct compact sets whose symmetric difference has area zero. The inequality follows by writing the area difference as $|E \setminus F| - |F \setminus E|$ and taking absolute values. The triangle inequality follows either from characteristic functions or $E \triangle G \subset (E \triangle F) \cup (F \triangle G)$.
- (b) The neighborhood of $[a, b]$ of radius ε is $[a - \varepsilon, b + \varepsilon]$. The two neighborhood inclusions are equivalent to $|a - c|, |b - d| \leq \varepsilon$. For the given intervals, $d_H = 1$ and $d_1 = 2$. (c) If $K \subset L$, the excess $\rho(K, L)$ is zero, even if some points of L are far from K , for example $K = \{0\}$, $L = [0, 100]$.

Exercise 7.2

The thin rectangle adds area $1/n$, whereas its farthest points have horizontal distance 1 from S . Thus $d_1(K_n, S) = 1/n$ and $d_H(K_n, S) = 1$. The grid has area zero. Each grid square has its farthest point at its center, at distance $\sqrt{2}/(2n)$ from the nearest grid vertex, so $d_1(G_n, S) = 1$ and $d_H(G_n, S) = \sqrt{2}/(2n)$.

The examples disprove both implications. In particular, Hausdorff convergence does not imply area continuity without additional assumptions. The actual Hausdorff limit of K_n is $S \cup ([1, 2] \times \{0\})$, at distance at most $1/n$ from K_n ; its area is still 1. The two limits are identical only as equivalence classes modulo null sets.

Exercise 7.3

Matching points with the same first coordinate bounds the Hausdorff distance by $1/n$; the upper endpoint of K_n realizes this distance from L , so equality holds. But $K_n \cap L = \{(0, 0)\}$ for all n , and its distance from $L \cap L = L$ is 1. The failure occurs despite convexity and nonempty intersections.

Exercise 7.4

- (a) For a nearest point $z \in K$ to y , the triangle inequality gives $d_K(x) \leq |x - y| + d_K(y)$; exchange x, y . (b) First approximate a point in one set by a point in the other, giving $d_K \leq d_L + d_H(K, L)$ and its reverse. At $x \in K$, $|d_K(x) - d_L(x)| = d_L(x)$; maximize over K , then over L . This is the supremum norm on continuous functions. An essential supremum relative to ambient Lebesgue measure would be inappropriate when D has measure zero.
- (c) If $x, y \in K$ realize its diameter, choose $x', y' \in L$ within $\varepsilon = d_H(K, L)$. Then $|x - y| \leq |x' - y'| + 2\varepsilon \leq \text{diam } L + 2\varepsilon$; exchange the sets. (d) For $x, y \in K$, approximate by $x_n, y_n \in K_n$. For each fixed $t \in [0, 1]$, convexity puts $z_n = (1 - t)x_n + ty_n$ in K_n . Its limit z has $d_K(z) = 0$. Closedness of K gives $z \in K$.

Exercise 7.5 and synthesis

The limit f is nonnegative and 1-Lipschitz. Compactness of D gives a convergent subsequence of $x_n \in K_n$; uniform convergence implies its limit is a zero of f . For any zero y , the Lipschitz bound gives $f(x) \leq |x - y|$, hence $f \leq d_K$. For fixed x , a convergent subsequence of nearest points y_n has limit y with $f(y) = 0$. Thus $f(x) = \lim |x - y_n| = |x - y| \geq d_K(x)$. The identity $f = d_K$ now holds at every point of D ; uniform convergence of the common function subsequence gives convergence of the sets. The point-dependent auxiliary subsequences establish an identity, not a new simultaneous choice of subsequence for all points.

Thin rectangles can converge to a segment, so interior need not survive. Unit-area disks translated to infinity escape every fixed container; even after translation, the unit-area rectangles $[0, n] \times [0, 1/n]$ have unbounded diameters. An existence proof needs confinement, admissibility of the limit, and the appropriate semicontinuity of the objective.

Optional compactness details

Arzelà–Ascoli in the needed setting. Choose a countable dense subset of compact D by taking the union of finite $1/m$ -nets. Uniform boundedness and repeated subsequence extraction, followed by a diagonal choice, give convergence at every point of this subset. Given $\varepsilon > 0$, equicontinuity supplies a radius making all function oscillations less than $\varepsilon/3$. Cover D with finitely many such balls centered in the dense subset. Convergence at their centers makes the diagonal sequence uniformly Cauchy by the triangle inequality. Completeness of $\mathbb{R}$ gives a uniform limit, which is continuous. This proves the version used in Session 7.

Decreasing sets. If K_n decrease in a compact container, their intersection K is nonempty by the finite-intersection property. Otherwise some $x_j \in K_{n_j}$ would stay at least $\varepsilon > 0$ from K . A convergent subsequence has limit in every fixed K_m , and hence in K , a contradiction. Therefore $d_H(K_n, K) \rightarrow 0$.

Connected limits. A disconnected compact limit has a partition $K = A \cup B$ into disjoint nonempty compact subsets with $\text{dist}(A, B) > 0$. Small disjoint neighborhoods of A and B cover every sufficiently close approximant, and the approximant meets both neighborhoods. Their relative intersections separate it, contradicting connectedness.

Upper semicontinuity of volume for compact sets. For each $\varepsilon > 0$, eventually $K_n \subset K_\varepsilon$, so $\limsup |K_n| \leq |K_\varepsilon|$. As $\varepsilon \downarrow 0$, the neighborhoods decrease to K and have finite measure; continuity of measure from above gives $\limsup |K_n| \leq |K|$. The grid example satisfies this inequality strictly. Thus area is upper semicontinuous but not continuous on general compact sets; the topology on open complements in Henrot–Pierre must be treated separately.

8 Lab 8: support functions and the minimum-width problem

Teaching choice. In the course, establish the support-function metric identity, explain the continuity mechanisms, and prove existence using area sublevels. Present Pál's identification through the incircle construction, supplying its contact and separation lemmas. The guided lab completes the area calculation and monotonicity argument. The full geometric justification below is instructor background, not extra work to squeeze into the 90-minute lecture.

Exercise 8.1

- (a) $h(\theta) = 2 \cos \theta + 1$, which equals -1 at $\theta = \pi$. The width is constantly 2; translation terms cancel.
 (b) Maximizing the two coordinate contributions independently gives $h_K(\theta) = a|\cos \theta| + b|\sin \theta|$. Its minimum is $\min(a, b)$: it is at least $\min(a, b)(|\cos \theta| + |\sin \theta|) \geq \min(a, b)$, with equality on a suitable coordinate axis. Cauchy–Schwarz gives the maximum $\sqrt{a^2 + b^2}$, attained in a direction proportional to (a, b) . Width is twice support here.
 (c) The radius- r circle and its filled disk centered at zero both have support function r , but their Hausdorff distance is r . The circle as a point set is not convex. (d) All three listed shapes have minimum width 1. The disk and Reuleaux triangle have constant width 1, while the equilateral triangle has maximum width $2/\sqrt{3} > 1$.

Exercise 8.2

- (a) The support differences are bounded by ε_n , so the width differences are bounded uniformly by $2\varepsilon_n$. The minima of two functions differ by at most their uniform distance. Uniform convergence also preserves equality to 1 in every direction. (b) Both sets lie inside $(1 + \varepsilon_n/r)K$ and contain $(1 - \varepsilon_n/r)K$, so their symmetric difference has area at most

$$[(1 + \varepsilon_n/r)^2 - (1 - \varepsilon_n/r)^2]A(K) = 4\varepsilon_n A(K)/r.$$

- (c) Area is $1/n$, perimeter $2 + 2/n$, and the limiting extended perimeter is 2. A planar compact convex set with empty interior is contained in a line and has zero minimum width.

Exercise 8.3

Rectangles $[0, n] \times [0, 1]$ show that width alone supplies no upper diameter bound. If $d = |AB|$ is a diameter, the support points above and below its line give two triangles inside K whose total area is $d(h_+ + h_-)/2 \geq dw(K)/2$. The equilateral competitor of altitude 1 has area $1/\sqrt{3}$. A minimizing sequence with $A(K_n) \leq 2/\sqrt{3}$ therefore has diameter at most $4/\sqrt{3}$. Translate a point of each set to the origin, preserving both area and width. Blaschke gives a convex limit in the closed ball of that radius; width continuity keeps $w \geq 1$, and area continuity gives attainment. If $w > 1$, dilation by $1/w$ would strictly decrease its positive area while preserving admissibility. Hence $w = 1$ at a minimizer.

Exercises 8.4–8.5

Tangency gives $PX = \sqrt{q^2 - r^2} = \sqrt{1 - 2r}$ and $\cos \alpha = r/q = r/(1 - r)$. The two triangles OPX have total area $r\sqrt{1 - 2r}$; the intervening disk sector has angle 2α and area $r^2\alpha$. Adding three nonoverlapping protrusions to the disk gives the stated $F(r)$. Overlap would invalidate this sum as a lower bound on the area of their union.

For $1/3 < r < 1/2$, $1/2 < r/(1-r) < 1$, so $0 < \arccos(r/(1-r)) < \pi/3$. Both terms in the supplied derivative are positive. At $r = 1/3$, the circular terms cancel and $F(1/3) = 1/\sqrt{3}$. If $r = 1/2$, the disk gives the stronger bound $\pi/4 > 1/\sqrt{3}$. Equality therefore requires $r = 1/3$ and hence an equilateral triangle. Scaling by w_0 gives area $w_0^2/\sqrt{3}$, altitude w_0 , and side length $2w_0/\sqrt{3}$. The triangle does not satisfy the constant-width constraint.

For the optional differentiation, write $s = \sqrt{1-2r}$ and $\alpha = \arccos(r/(1-r))$. Then $s' = -1/s$ and $\alpha' = -1/((1-r)s)$. Substitution gives

$$F' = 2\pi r - 6r\alpha + 3s - \frac{3r}{s} + \frac{3r^2}{(1-r)s} = 6r(\pi/3 - \alpha) + \frac{3s^3}{1-r}.$$

Support-function consequences: supplementary proofs

Area and integrals. The course proves L^1 convergence when the limit contains a disk. If the limit is a segment of length ℓ , its ε -neighborhood has area $2\ell\varepsilon + \pi\varepsilon^2$; for a point use $\pi\varepsilon^2$. Thus area convergence also holds at degenerate limits. For bounded continuous f on the common container,

$$\left| \int_{K_n} f - \int_K f \right| \leq \|f\|_\infty |K_n \triangle K| \rightarrow 0.$$

Cauchy's formula without presupposing perimeter continuity. For a convex polygon with side vectors $\ell_i v_i$, projection onto u_θ gives $\sum_i \ell_i |v_i \cdot u_\theta| = 2w_K(\theta)$: the two boundary chains each traverse the projected interval monotonically. Integrating on $[0, \pi]$ uses $\int_0^\pi |v_i \cdot u_\theta| d\theta = 2$ and gives $P(K) = \int_0^\pi w_K(\theta) d\theta$. For a convex body, perimeter is the supremum of lengths of inscribed polygons. Choose polygons whose perimeters approach this supremum, and add increasingly dense finite boundary nets to their vertices. Adding vertices does not reduce perimeter; the resulting polygons converge in Hausdorff distance. Their widths converge uniformly, so the formula passes to the limit. In particular the supremum is finite, since every inscribed polygon has perimeter at most $\pi \text{diam } K$. This proves the formula before using it to establish perimeter continuity. For a segment, direct integration gives twice its length. Constant width w_0 gives $P = \pi w_0$ (Barbier's theorem).

Inradius and circumradius. The largest radius of a ball centered at c and contained in K is controlled by

$$g_K(c) = \min_{|u|=1} (h_K(u) - c \cdot u), \quad r(K) = \max_{c \in \mathbb{R}^d} g_K(c).$$

The maximum is attained because g_K is continuous and tends to $-\infty$ as $|c| \rightarrow \infty$; its maximum is nonnegative by choosing $c \in K$. Support reconstruction shows that a nonnegative value is exactly the admissible radius at that center. Uniform support differences at most ε give $|g_K(c) - g_L(c)| \leq \varepsilon$ for every c , hence $|r(K) - r(L)| \leq \varepsilon$. Similarly,

$$R(K) = \min_c \max_{x \in K} |x - c|.$$

For each fixed c , the two inner maxima differ by at most $d_H(K, L)$; taking minima proves the circumradius bound. The latter argument does not require convexity.

Full geometric justification of the incircle proof

The following develops the manuscript's construction. Yaglom–Boltyanskii, Chapter 6, Exercise 6-2 and its solution give the contact geometry; Exercise 6-4 and its solution give the minimum-width theorem. The one-variable calculation is used here in place of the book's final geometric comparison.

1. Contact normals surround the center. Normalize $w(K) = 1$ and translate a largest inscribed disk to $B = \overline{B}(0, r)$. Existence of this disk follows from the inradius formula above; positive width in the plane gives interior and hence $r > 0$. Let $N = \{u : |u| = 1, h_K(u) = r\}$ be its active contact normals. We claim $0 \in \text{conv } N$. Otherwise strict separation supplies v and $\eta > 0$ with $u \cdot v \geq \eta$ for all $u \in N$. Move the center to $-tv$. In a fixed neighborhood of N , all supporting-plane clearances increase by at least $t\eta/2$. On the compact remainder of the unit circle, the original clearance exceeds r by a positive uniform amount. Taking $t > 0$ sufficiently small therefore permits a disk of radius greater than r , a contradiction.

In the plane, a point in a convex hull is in the convex hull of at most three generating points: in a convex combination of four or more points, affine dependence lets one change the coefficients until one vanishes, then repeat. Thus there are either two antipodal contact normals or three whose triangle contains zero in its interior. In the three-normal case the successive angular gaps are all less than π , and their contact tangent lines enclose a triangle $T \supset K$ with the same incircle.

2. The radius bounds and their equality cases. Since the disk is inside K , $2r \leq w(K) = 1$. If two contacts are antipodal, K is confined to their strip of width $2r$. Hence $1 = w(K) \leq 2r$, giving $r = 1/2$ and $A(K) \geq \pi/4$. Otherwise write the sides of the contact triangle as a, b, c , with a longest. Its smallest altitude is

$$w(T) = \frac{2A(T)}{a} = \frac{r(a+b+c)}{a} \leq 3r.$$

Here the minimum width of a triangle is its smallest altitude. One quick justification is that between directions normal to edges, the two extreme vertices are fixed and their projection difference has second derivative equal to its negative. It is positive and strictly concave there, so a minimum occurs at an edge-normal direction. Inclusion gives $1 = w(K) \leq w(T) \leq 3r$, hence $r \geq 1/3$.

If $r = 1/3$, equality forces $a = b = c$, so T is equilateral of altitude 1. In each of its three side-normal directions, K touches the side and has width at least 1. It must therefore contain the opposite vertex of T , the only point of T at distance 1 from that side. Thus K contains all three vertices, and $K = T$ by convexity and inclusion.

3. Find three radial endpoints inside K . Assume $1/3 < r < 1/2$ and put $q = 1 - r > r$. Label the three contact normals u_1, u_2, u_3 cyclically. The supporting lines are $u_i \cdot x = r$. Choose a support point Q_i on the opposite side. Since the directional width is at least 1,

$$u_i \cdot Q_i = r - w_K(u_i) \leq r - 1 = -q, \quad |Q_i| \geq q.$$

Consequently $P_i = qQ_i/|Q_i|$ belongs to K , because the segment from the center to Q_i lies in K . Each $H_i = \text{conv}(B \cup \{P_i\})$ is contained in K and has the same area, independent of its angular position.

4. The added regions do not overlap. The three sectors between consecutive rays u_1, u_2, u_3 partition the plane. We show that $H_i \setminus B$ lies in the sector between the other two contact rays, namely the sector not containing u_i .

First, Q_i cannot be in a sector generated by u_i and u_j . For any vector $x = su_i + tu_j$ with $s, t \geq 0$,

$$(u_i + u_j) \cdot x = (s + t)(1 + u_i \cdot u_j) \geq 0.$$

But $u_i \cdot Q_i \leq -q$ and $u_j \cdot Q_i \leq r$, whose sum is negative because $q > r$. This excludes both sectors adjacent to u_i , so Q_i , and hence P_i , is in the opposite sector.

For a point P_i at radius q , the visible disk arc between its two tangencies has angular half-width $\alpha = \arccos(r/q)$ about the ray OP_i . The constraints $u_j \cdot P_i \leq r$ for the two bounding contact normals mean that neither bounding ray lies in the interior of this visible angular interval. Since the ray OP_i is inside the sector, its entire visible interval, and the added tangent region, stays in that sector. The three added regions therefore have disjoint interiors. It suffices to use the union $B \cup H_1 \cup H_2 \cup H_3 \subset K$; no assertion about the area of their full convex hull is needed.

5. Add areas and conclude. The tangent calculation in Exercise 8.4 now applies to each region, giving $A(K) \geq F(r)$. The strict monotonicity in Exercise 8.5 and the endpoint cases above prove the inequality and characterize equality. No optimizer was assumed in deriving this universal bound. The earlier direct-method argument remains valuable as an independent existence proof and as a model for problems where the optimizer cannot be explicitly identified.

Optional: why the Reuleaux triangle has constant width

Set $A = (0, 0)$, $B = (1, 0)$, $C = (1/2, \sqrt{3}/2)$ and intersect the three closed unit disks centered at these points. On the boundary arc from B to C centered at A , the outward normal u ranges from angle 0 to $\pi/3$, and the supporting value is $h_K(u) = 1$. The opposite support point is A : the disk constraints at A give $x \cdot B \geq 0$ and $x \cdot C \geq 0$ for every $x \in K$, and u is a nonnegative combination of B and C . Hence $\min_{x \in K} x \cdot u = 0$, giving width 1. Rotate this argument by $2\pi/3$ and $4\pi/3$ for the other two arcs. Together with $w_K(-u) = w_K(u)$, these intervals cover every direction. Scaling gives the result for any width w_0 .

9 Lab 9: mixed areas, boxes and the isoperimetric bound

Teaching choice. The box induction is the main new proof. Use the plane for drawings but retain d in the volume calculation: the argument does not require induction on dimension. Prove the planar Steiner formula geometrically and state the general volume-polynomial theorem. Keep mixed area measures, curvature measures and the Alexandrov–Fenchel inequality outside the core.

Exercise 9.1

The sum is $[0, 3] \times [0, 4]$, and $A(K + tL) = (2 + t)(1 + 3t) = 2 + 7t + 3t^2$. Hence $V(K, L) = 7/2$, whereas $A(K \cap L) = 1$. For any unit u ,

$$\max_{x \in K, y \in L} (x + y) \cdot u = \max_{x \in K} x \cdot u + \max_{y \in L} y \cdot u.$$

Convexity gives $K + K = 2K$, so the area polynomial at $s = t = 1$ gives $4A(K) = 2A(K) + 2V(K, K)$. Reflection gives $h_{-K}(u) = h_K(-u)$, so $h_{K+(-K)} = w_K$. A constant support value w identifies the closed ball centered at zero of radius w by support reconstruction.

Exercise 9.2

The strips total $6t$, and the four quarter-disks total πt^2 . Thus $A(K + tB) = 2 + 6t + \pi t^2$ and $P(K + tB) = 6 + 2\pi t$. At $t = 1/2$, these are $5 + \pi/4$ and $6 + \pi$. The linear area coefficient is 6, so $V(K, B) = 3$. To first order, a uniform outward motion of each side adds its length times the displacement; corners contribute only a quadratic term.

For the separated disks, the parallel disks have radius $1 + t$ and are still disjoint for the stated range. Their union has area $2\pi(1 + t)^2 = 2\pi + 4\pi t + 2\pi t^2$. The convex formula cannot be applied to this disconnected union.

Exercise 9.3

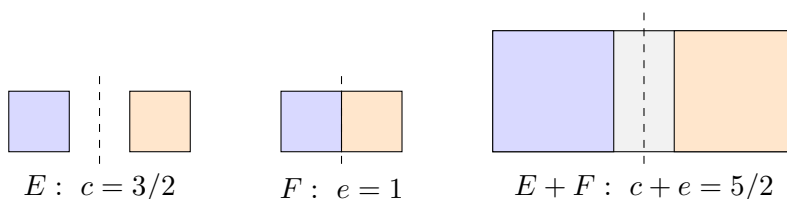
The left fraction of E is $1/2$, so cut F at $x = 1$. The pieces are

$$E^- = [0, 1] \times [0, 1], \quad E^+ = [2, 3] \times [0, 1], \quad F^- = [0, 1]^2, \quad F^+ = [1, 2] \times [0, 1].$$

Their matching sums are $[0, 2] \times [0, 2]$ and $[3, 5] \times [0, 2]$, of area 4 each. They lie on opposite sides of $x = 5/2$, the sum of the two cutting coordinates. Since both original areas are 2, the lower bound is $(\sqrt{2} + \sqrt{2})^2 = 8$. But the full sum is

$$([0, 3] \times [0, 2]) \cup ([2, 5] \times [0, 2]) = [0, 5] \times [0, 2],$$

of area 10. The omitted band illustrates why inclusion, not equality, is the correct set relation in the proof.



For the abstract count, choose two boxes of K separated by the cut. Each side misses the interior of at least one selected box, so each uses at most $m - 1$ boxes. Every original box of L contributes at most one box to either specified side, so each side uses at most n . Thus each recursive pair has at most $m + n - 1$ boxes. Counting all four pieces together is irrelevant and can indeed give a larger number.

Exercise 9.4

(a) Addition gives $E + F \subset E_j + F_j \subset E + F + 2\delta_j B$. Since $E + F$ is compact, its shrinking closed neighborhoods have volumes converging to $|E + F|$ by continuity from above. Squeezing yields the desired convergence, and similarly for E_j, F_j separately. Mere Hausdorff convergence would not suffice: finite grids converging to a square have area zero throughout.

(b) Cancel the constant and quadratic terms in the two expansions and divide by $2t > 0$, obtaining $V(K, L) \geq \sqrt{A(K)A(L)}$. The rectangle check is $49/4 \geq 6$, strictly. Equality of the mixed-area bound forces equality of the full area polynomial with the Brunn–Minkowski lower bound for every $t > 0$. The supplied convex-body rigidity theorem then gives homothety. For aligned rectangles this reduces to matching side ratios, which can also be checked directly from $(ad - bc)^2 \geq 0$.

(c) Use $V(K, B) = P(K)/2$ and $A(B) = \pi$. For the directional perimeter, the lower bound is $P_L(K) \geq 2\sqrt{A_0 A(L)}$. It is attained by $K = \sqrt{A_0/A(L)} L + z$. Homothety gives all equality shapes under the stated positive-area assumptions. No existence theorem is needed for this last model because an attaining competitor is explicitly given.

Exercise 9.5

Here

$$|K + tL| = (1 + 2t)(1 + 3t)(1 + 4t) = 1 + 9t + 26t^2 + 24t^3.$$

Thus $V(K, K, L) = 3$ and $V(K, L, L) = 26/3$, rather than 9 and 26. Minkowski's first inequality gives $3^3 = 27 \geq 24$, strictly because the two boxes are not homothetic. Cubing $S \geq 3(4\pi/3)^{1/3}|K|^{2/3}$ gives $S^3 \geq 36\pi|K|^2$.

Optional isodiametric application

Every $x - y \in K + (-K)$ has norm at most D , so the difference body is inside DB . Brunn–Minkowski and reflection invariance give

$$2^d|K| \leq |K + (-K)| \leq \kappa_d D^d.$$

For constant width w , diameter is w , and in the plane $A(K) \leq \pi w^2/4$, attained by a disk of radius $w/2$. This identifies the area maximum in the constant-width class whose existence was proved in Session 8. It does not identify its area minimum.

If equality holds in the isodiametric bound for a convex body of positive volume, equality holds for K and $-K$ in Brunn–Minkowski. The supplied rigidity theorem makes them homothetic; equal volumes make their scale factor one. Thus K is centrally symmetric up to translation. Equality of volumes in the inclusion $K + (-K) \subset DB$ forces equality of these convex bodies, so K is a ball of radius $D/2$.

Supplementary proof details for the lecture

1. Why the polygon polynomial is valid. For each polygon, traverse its boundary counterclockwise in increasing edge direction. Merge the two direction lists. When one polygon has no edge in a listed direction, its support vertex stays fixed for that step; represent that step by a zero edge. Starting with support vertices for one common normal direction fixes compatible cyclic lists p_i, q_i . Exposed faces add under Minkowski addition, so the sum polygon has vertices $sp_i + tq_i$ (possibly with repetitions when a coefficient is zero). The shoelace expression becomes

$$\frac{1}{2} \sum_i \det(sp_i + tq_i, sp_{i+1} + tq_{i+1}),$$

which is homogeneous quadratic. The pure coefficients are the respective polygon areas, and the remaining coefficient is $2V(K, L)$. Merging the edge lists of three polygons at once shows that this cross term is linear in each polygon's vertex list; hence it is Minkowski additive in each argument.

For convex-body limits $P_j \rightarrow K, Q_j \rightarrow L$, continuity gives

$$V(P_j, Q_j) = \frac{1}{2}(A(P_j + Q_j) - A(P_j) - A(Q_j)) \longrightarrow \frac{1}{2}(A(K + L) - A(K) - A(L)).$$

Pass to the limit in the polynomial identity for each fixed s, t . This also proves continuity of mixed area and passes additivity to the limit. No area-continuity assertion for nonconvex approximants is used.

2. The edge first-variation formula. For a fixed polygon K , first take L polygonal and translate it so $0 \in L$. The common edge-direction description shows that the support line normal to each edge of K moves by $th_L(u_i)$. Away from its endpoints, the added strip has area $t\ell_i h_L(u_i)$; endpoint changes and any new edges from L stay within $O(t)$ of vertices of K , giving only $O(t^2)$ area. There are finitely many vertices, so

$$A(K + tL) - A(K) = t \sum_i \ell_i h_L(u_i) + O(t^2).$$

Compare derivatives with the quadratic polynomial. For general convex L , approximate by polygons and use uniform support convergence and continuity of V to obtain the exact edge formula. Translating L by z changes the right-hand side by $z \cdot \sum_i \ell_i u_i = 0$; the vector sum vanishes because the edge vectors of a closed polygon sum to zero and normal rotation is linear. Monotonicity follows from support order and passes to general K by convex polygonal approximation. In particular $V(K, L) \geq 0$, by translating L to contain the singleton $\{0\}$, whose mixed area is zero.

3. Existence of a matching cut. For a box $R = \prod_k [a_k, b_k]$, its volume to the left of $x_i = e$ is

$$\min\{\max\{e - a_i, 0\}, b_i - a_i\} \prod_{k \neq i} (b_k - a_k).$$

This is continuous in e . For a union with disjoint interiors, add these functions. The total runs continuously from 0 to $|L|$, so the intermediate value theorem supplies every fraction $\theta \in (0, 1)$. Strict monotonicity is unnecessary: gaps between boxes can produce plateaus, but they do not prevent existence of the desired level.

4. The count and the approximation. Two axis-aligned boxes with disjoint interiors have interval interiors disjoint in at least one coordinate; otherwise their product interiors would meet. Cutting between those two intervals ensures that the two selected boxes are not split through their interiors. Other boxes may be split. Define the sets on each side using only clipped boxes of positive volume, and use the same convention for L . This preserves their individual volumes and

their containment in the original sets. It need not preserve a Minkowski sum's volume: a null set can have a positive-volume sum with another set. No such preservation is needed, because the proof uses only the sums of the retained pieces as subsets of $K + L$. Those sums may overlap on their separating plane, but that plane has zero d -volume.

For outer approximations of a nonempty compact set, use the finite union of all closed dyadic cubes meeting it. Interiors of grid cubes are disjoint, and any point of a selected cube is within its diameter $\sqrt{d}2^{-j}$ of the compact set. The neighborhood squeeze in the lecture gives volume convergence even for compact sets with empty interior or positive-volume sets with irregular boundaries. The sum is compact, so the same measure argument applies to it. This avoids assuming either boundary regularity or general Hausdorff continuity of volume.

5. Higher-dimensional equality after differentiation. For positive-volume convex bodies, put $f(t) = |K + tL|^{1/d}$. Brunn–Minkowski makes f concave on $[0, \infty)$, and the volume polynomial gives its right derivative at zero. Hence

$$f(0) + t|L|^{1/d} \leq f(t) \leq f(0) + tf'(0+).$$

The upper bound is the tangent-line inequality for a concave function. If equality holds in Minkowski's first inequality, then $f'(0+) = |L|^{1/d}$, so both bounds coincide for every t . The stated Brunn–Minkowski rigidity theorem now gives homothety. With $L = B$, this is the ball equality case. Simply differentiating an inequality would not, by itself, justify this equality conclusion.

Reference conventions and scope. Schneider's local filename contains 2013, but the title/copyright pages identify the Second Expanded Edition as 2014. The volume-polynomial theorem is Theorem 5.1.7; the inequality and convex equality conditions are Theorems 7.1.1 and 7.2.1. The box-cut proof is in Treibergs, Section 8, and is discussed by Schneider in the notes to Section 7.1, item 10 (printed p. 379), with a reference to Hadwiger–Ohmann and Gardner's survey. We use the compact-set version of the inequality and state rigidity only for positive-volume convex bodies. We use only the convex Steiner formula: the unrestricted disconnected-set version with a single πt^2 term would be false.

10 Lab 10: constant width and the mixed-area proof

Teaching choice. Begin with a caliper drawing and the Reuleaux triangle, then reuse the support and mixed-area tools already established. Prove the hexagon lemma rather than hiding it inside the short proof. The support-function differential formulas are needed for the Python supplement, but their general nonsmooth theory is outside the core.

Exercise 10.1

The support of $K + (-K)$ is $h_K(u) + h_K(-u)$. A constant value w identifies wB by support reconstruction. For a nonconvex set, the support function identifies only its convex hull, so the same reasoning cannot identify the original set. The width-one disk is $B/2$ and its difference body is B . A centrally symmetric body, translated to its center, satisfies $K = -K$, whence $2K = wB$.

Minkowski interpolation gives width $(1-t) + t = 1$. Its area is $(1-t)^2 A(K) + 2t(1-t)V(K, L) + t^2 A(L)$, usually not linear. For example, interpolate a centered Reuleaux triangle with its negative. The midpoint is $B/2$, whose area exceeds that of either endpoint.

Exercise 10.2

The width sum is one, $h + h'' = 1/2 - 8a \cos 3\theta$, and feasibility is exactly $|a| \leq 1/16$. With $h' = -3a \sin 3\theta$, the coordinates are

$$x_1 = (1/2 + a \cos 3\theta) \cos \theta + 3a \sin 3\theta \sin \theta, \quad x_2 = (1/2 + a \cos 3\theta) \sin \theta - 3a \sin 3\theta \cos \theta.$$

Integrating squares gives $A = \pi/4 - 4\pi a^2$, while $P = \pi$. The family minimum is $15\pi/64 \simeq 0.736311$, attained at $a = \pm 1/16$; these two bodies differ by a rotation. It exceeds $(\pi - \sqrt{3})/2 \simeq 0.704771$, so the finite family excludes the global minimizer. Adding $b \cos \theta + c \sin \theta$ adds (b, c) to the boundary; the curvature radius, width, perimeter and area are unchanged.

Why the smooth convexity criterion works. For $x(\theta) = hu + h'u^\perp$, differentiation gives $x' = \rho u^\perp$. Fix α . The derivative of $x(\theta) \cdot u(\alpha)$ is $\rho(\theta) \sin(\alpha - \theta)$. It is nonpositive on $[\alpha, \alpha + \pi]$ and nonnegative on $[\alpha + \pi, \alpha + 2\pi]$. By periodicity its global maximum is its value $h(\alpha)$ at $\theta = \alpha$. Thus every $x(\theta)$ lies in all the supporting half-planes, with equality at the appropriate normal, and the convex hull of this curve has support exactly h . This proves sufficiency even if ρ vanishes. Necessity follows from the nonnegative speed in the convex tangent orientation. For $\rho > 0$, Green's formula $A = \frac{1}{2} \int \det(x, x')$ gives $A = \frac{1}{2} \int h\rho$ and integration by parts gives the displayed area formula. Add a small constant to h and pass to the limit to cover zeros of ρ .

For a finite odd Fourier expansion with frequency one removed, orthogonality gives the useful general identity

$$A = \frac{\pi w^2}{4} - \frac{\pi}{2} \sum_{j \geq 3, j \text{ odd}} (j^2 - 1)(a_j^2 + b_j^2).$$

The negative quadratic terms encourage large coefficients, but convexity couples those coefficients through the pointwise inequality $h + h'' \geq 0$. A sufficient, conservative certificate is $\sum_j (j^2 - 1)\sqrt{a_j^2 + b_j^2} \leq w/2$. A finite grid test of $\rho \geq 0$ alone is not a continuous certificate.

Exercise 10.3

Translate a point of each member of the sequence to zero. Diameter one implies containment in the unit disk, Blaschke gives a convex Hausdorff limit, and uniform support convergence

preserves the width identity. The limit has positive width in every direction and hence nonempty interior. Convex area continuity proves attainment.

The sign-change identity is $F(\theta + \pi/3) = m(\theta + \pi/3) - m(\theta + 2\pi/3) - m(\theta) = -F(\theta)$. At a zero, use $u_0 - u_1 + u_2 = 0$ to obtain the third equation from the first two. Since u_0, u_1 are independent, c is well defined. The six support lines are $x \cdot u_j = c \cdot u_j \pm 1/2$. Their regular hexagon has side $1/\sqrt{3}$, perimeter $2\sqrt{3}$, area $\sqrt{3}/2$, and circumradius $1/\sqrt{3}$.

Exercise 10.4

Place T on the circle of radius $1/\sqrt{3}$ with vertex angles $0, 2\pi/3, 4\pi/3$. The six vertices of $T \cup (-T)$ form a regular hexagon. The three unit arcs stay in this hexagon and touch its sides; therefore $H = \text{conv}(T \cup (-T)) = \text{conv}(R \cup (-R))$. The areas are respectively $\sqrt{3}/4$, $(\pi - \sqrt{3})/2$ and $\sqrt{3}/2$. In $T + (-T)$ the six nonzero differences have length one and are regular hexagon vertices. Its area is $3\sqrt{3}/2$, giving $V(T, -T) = \sqrt{3}/2$. For $T \subset L \subset H$, reflection preserves inclusions and monotonicity traps $V(L, -L)$ between this value and $V(H, H) = \sqrt{3}/2$.

For arbitrary width-one K , use the covering lemma to place $K, -K \subset H$. Then $A(K) = \pi/2 - V(K, -K) \geq \pi/2 - \sqrt{3}/2$. The Reuleaux triangle attains this. Strict inclusion in the sandwich example coexists with equality of mixed areas, so no strict-monotonicity argument is available for uniqueness. Barbier and isoperimetry give $A(K) \leq \pi/4$, with disk equality.

Synthesis. At minimum width one, the least area is $1/\sqrt{3}$ and the minimizer is an equilateral triangle of altitude one. Its directional widths are not all one. At constant width one, the least area is $(\pi - \sqrt{3})/2$ and the minimizer is a Reuleaux triangle. The former class requires an area-sublevel diameter bound; the latter has diameter one automatically.

Optional: the reverse isoperimetric viewpoint

Here is one precise extension from Bogosel (2023), Section 3. We *supply* the planar convex-hull identity

$$A(S) = V(S, K_1) + V(S, K_2) - V(K_1, K_2), \quad S = \text{conv}(K_1 \cup K_2).$$

Its proof uses additional mixed-area theory, so it is not a prerequisite for either course proof. For $K_1 = K$, $K_2 = -K$ and width one, bilinearity and $K + (-K) = B$ give

$$V(K, -K) = V(S, B) - A(S) = P(S)/2 - A(S).$$

Thus the original minimization is equivalent to maximizing this quantity over hulls S that actually arise from constant-width bodies. Their supports satisfy $h_S = \max(h_K, 1 - h_K) \geq 1/2$. Center K at the center of a covering hexagon. Since the hexagon has circumradius $1/\sqrt{3}$, we obtain

$$(1/2)B \subset S \subset (1/\sqrt{3})B.$$

One may relax to all convex bodies between these two disks. The cited paper uses a separate annulus-constrained optimization theorem to solve that relaxation by the regular hexagon; we do not reprove that theorem. Both radii matter: with only the inner condition, the unit square has $P/2 - A = 1 > \sqrt{3}/2$; with only the outer condition, a diameter segment has extended perimeter $4/\sqrt{3}$ and objective $2/\sqrt{3} > \sqrt{3}/2$.

Three dimensions: what the coefficients do prove. Write $v = |K|$ and $a = V(K, K, -K)$ for a width-one body in $\mathbb{R}^3$. Reflection and symmetry imply $V(K, -K, -K) = a$. The volume polynomial gives

$$4\pi/3 = 2v + 6a, \quad S(K)/3 = V(K, K, B) = v + a.$$

Elimination yields Blaschke's identity $v = S(K)/2 - \pi/3$. At width w , scaling gives $v = wS(K)/2 - \pi w^3/3$. Existence follows as before, and minimizing volume is equivalent to minimizing surface area. This does not identify the optimizer. The cited papers discuss the Meissner-body conjecture and the difficulty of finding a sufficiently sharp covering body. The ordinary Reuleaux tetrahedron is not a constant-width body. No three-dimensional Blaschke–Lebesgue assertion is used in the course.

11 Lab 11: tangent polygons, perturbations and multipliers

Teaching choice. Spend most of the lecture on the paired-cut proof and the geometry of an admissible motion. Derive the gradient in an optional board extension or use the derivation below as reading. Students can complete the mathematical lab using the supplied gradient.

Exercise 11.1

The two adjacent outer normals are separated by δ ; their intersection angle inside the rhombus is $\pi - \delta$. Each removed triangle has half-base $t \tan((\pi - \delta)/2) = t \cot(\delta/2)$, so its area is $t^2 \cot(\delta/2)$. Writing $t_2 = s - t_1$ proves the bounds, with the lower equality at $t_1 = t_2$ and the upper equality at $t_1 t_2 = 0$. The perimeter loss for one triangle is $2t(\csc(\delta/2) - \cot(\delta/2))$, a linear function of t .

For $w = 1$, $\delta = \pi/3$, $s = 2/\sqrt{3} - 1$. There are three opposite pairs, so the total losses are

$$D_{\text{disk}} = \frac{3\sqrt{3}}{2}(2/\sqrt{3} - 1)^2, \quad D_R = 3\sqrt{3}(2/\sqrt{3} - 1)^2.$$

Numerically they are approximately 0.062178 and 0.124356. Subtracting from $\sqrt{3}/2$ gives the respective 12-direction supporting-polygon areas, about 0.803848 and 0.741669.

Exercise 11.2

Denote total refinement loss by $D_N(K) = A(Q_N(K)) - A(Q_{2N}(K))$. The local comparison gives $D_N(D_w) \leq D_N(K) \leq D_N(R)$. Consequently each of the gaps $A(Q_N(K)) - A(Q_N(R))$ and $A(Q_N(D_w)) - A(Q_N(K))$ is nondecreasing with refinement. Both start at zero. Area continuity gives the limiting bounds and preserves every positive finite-stage gap. By contrast, a sequence of strict inequalities $b_j > a_j$ can converge to equality; the numerical example $b_j = 1/j$, $a_j = 0$ suffices.

At a corner p , opposite contacts $p - wu$ trace a circular arc as u ranges over the normal cone. Its endpoint distance is bounded by the diameter w . Thus the normal-cone angle is at most $\pi/3$. If it equals $\pi/3$, the corner and the arc endpoints form an equilateral triangle of side w . Each pair of its vertices realizes the diameter, so the support lines perpendicular to the two incident diameters at each vertex are supporting lines. They give a normal cone of angle at least $\pi/3$, hence exactly $\pi/3$. The three opposite circular arcs therefore all belong to the boundary; they form the entire Reuleaux triangle boundary. Equality in the first refinement forces a hexagon vertex to be a support contact of K , producing just such a corner. This supplies lower equality rigidity.

Details of the tangent construction. The intersection point of supports h_j, h_{j+1} , expressed in the basis $(u_j, u_j^\perp)$, is

$$z_j = h_j u_j + \frac{h_{j+1} - h_j \cos \delta}{\sin \delta} u_j^\perp.$$

Subtract the analogous previous vertex to obtain the edge-length formula in the lecture. The perimeter identity follows by summing. In a refinement the old and new contact points occur in their cyclic normal order. Each new support cuts just the corresponding old corner; zero sides are interpreted by a limiting construction. The triangular cuts have disjoint interiors. This is the geometric justification for adding their areas, including at the degenerate polygons around R .

Exercise 11.3

The longest chord of the regular odd polygon subtends angle $\pi - \pi/n$ at its center, and has length $2r_n \cos(\pi/(2n)) = 1$. Its skeleton area is $nr_n^2 \sin(2\pi/n)/2$. Each added circular segment has area $(\pi/n - \sin(\pi/n))/2$. Adding and simplifying gives $A_n = \pi/2 - (n/2) \tan(\pi/(2n))$. Put $t = \pi/(2n)$ and differentiate:

$$\frac{dA_n}{dn} = \frac{1}{2}(t \sec^2 t - \tan t) > 0.$$

To check the sign, differentiate $t \sec^2 t - \tan t$ with respect to t : its derivative is $2t \sec^2 t \tan t > 0$ and its value at zero is zero. Finally $n \tan(\pi/(2n)) \rightarrow \pi/2$. This comparison identifies the best regular example only. One must first obtain a minimizer in a compact class, then show that every nondegenerate stationary example is regular, and finally pass from Reuleaux polygons to arbitrary constant-width bodies by approximation.

Exercise 11.4 and the multiplier extension

For $g_0 \neq 0$ take $v = -g_0$, giving $A'(0) = -|g_0|^2 < 0$. The dependent vertices preserve the unit diameters. Positivity of arc lengths and the cyclic arrangement are open conditions, so this gives an admissible descent for sufficiently small steps. At a stationary vertex the two independent diameter vectors force $\theta_0 = \theta_k = \theta_{k+1}$. In particular, stationarity at all vertices gives $\theta_j = \theta_{j+k}$; since $\gcd(k, 2k+1) = 1$, all angles coincide. Equal arc angles and equal corner turns imply that the skeleton is regular.

For a regular pentagon, $A_5 \simeq 0.758497 > A_3 \simeq 0.704771$. This global comparison alone does not exclude a local minimum. The supplied second derivative is approximately -1.453085 , so the specified local deformation does exclude one. In general, with $0 < \theta \leq \pi/5$, its numerator is negative because

$$\sin(\theta/2) \sin \theta - \cos(3\theta/2) = 2 \sin(\theta/2) \sin \theta - \cos(\theta/2) \cos \theta < 0.$$

Equivalently $\tan(\theta/2) \tan \theta < 1/2$; the product increases with θ and is less than $1/2$ at $\pi/5$. The formula is for a straight movement of x_0 along its inward unit bisector, with the two opposite vertices adjusted.

The class with exactly N distinct vertices is not closed: arcs can shrink and vertices merge. The actual limit has fewer vertices, still odd after redundant disks are removed. This is why existence is stated for at most N vertices.

Deriving the area gradient

For a moving boundary, the area derivative is the integral of outward normal velocity. For a circular arc with center velocity c' , half-angle $\beta/2$, and unit bisector b , its contribution is $2 \sin(\beta/2) c' \cdot b$; integrate $\cos t$ between $-\beta/2$ and $\beta/2$. Tangential velocities do not contribute.

More generally, suppose the endpoints a, b of a unit arc move with velocities v_a, v_b , while its center is c . Differentiating $|a - c| = |b - c| = 1$ gives

$$(a - c) \cdot c' = (a - c) \cdot v_a, \quad (b - c) \cdot c' = (b - c) \cdot v_b.$$

Since $a - c + b - c = 2 \cos(\beta/2) b_{\text{arc}}$, the arc contribution is

$$\tan(\beta/2)((a - c) \cdot v_a + (b - c) \cdot v_b).$$

Now move x_0 with velocity v . Only three arcs have moving centers and nonzero normal velocity. The arc centered at x_0 contributes $2 \sin(\theta_0/2)v \cdot b_0$. The adjacent arcs centered at x_k and x_{k+1} have x_0 as one endpoint and a fixed other endpoint; their contributions are $\tan(\theta_k/2)(x_0 - x_k) \cdot v$ and $\tan(\theta_{k+1}/2)(x_0 - x_{k+1}) \cdot v$. Finally

$$b_0 = \frac{x_k + x_{k+1} - 2x_0}{2 \cos(\theta_0/2)}.$$

Substitution gives the lecture's gradient. Notice that other arcs may gain or lose tangential length as endpoints move, but their fixed centers give zero normal velocity; endpoint terms cancel in the closed-boundary area derivative.

A complete finite-dimensional multiplier argument

In a neighborhood of a nondegenerate Reuleaux polygon, allow its vertices to move independently and form the convex disk-polygon by joining successive vertices with the appropriate short unit arcs. Its area is the smooth local function

$$D(x) = \frac{1}{2} \sum_j \det(x_j, x_{j+1}) + \sum_j f(|x_j - x_{j+1}|), \quad f(s) = \arcsin(s/2) - \frac{s}{2} \sqrt{1 - s^2/4}.$$

Impose $C_i = (|x_i - x_{i+k}|^2 - 1)/2 = 0$. Keep the cyclic convex order and the remaining pair distances at most one. At a nondegenerate nontriangle all nonprescribed pair-distance inequalities are strict: an extra diameter between vertices would force an arc endpoint coincidence or a 60° arc, the triangle case. Thus only the C_i are active in this local calculation.

The gradient of C_i has blocks $x_i - x_{i+k}$ at i , $x_{i+k} - x_i$ at $i + k$, and zeros elsewhere. These n gradients are independent. Indeed, if $\sum_i \mu_i \nabla C_i = 0$, the block at j reads

$$\mu_j(x_j - x_{j+k}) + \mu_{j-k}(x_j - x_{j-k}) = 0.$$

The two nonparallel vectors force both coefficients to vanish. The squared constraints have the same gradients as the distance constraints on the feasible set because the relevant distances are one.

Apply the arc calculation to D at a Reuleaux polygon:

$$\partial_{x_j} D = \tan(\theta_{j+k}/2)(x_j - x_{j+k}) + \tan(\theta_{j-k}/2)(x_j - x_{j-k}).$$

Session 4's multiplier theorem for $\mathcal{L} = D + \sum_i \lambda_i C_i$ now gives

$$\lambda_j = -\tan(\theta_{j+k}/2), \quad \lambda_{j-k} = -\tan(\theta_{j-k}/2).$$

Reindexing the second identity also gives $\lambda_j = -\tan(\theta_j/2)$. Thus $\theta_{j+k} = \theta_j$, all angles are equal, and the candidate is regular. Compare A_n with A_3 and use the compactness and density arguments to finish the variational proof of the inequality. At collapsed configurations, first remove redundant vertices; do not apply this rank argument to a singular list with repeated entries.

Compactness and approximation details

For compactness with at most N disks, pass to a subsequence with a fixed number of centers and then make every center converge. Centers belong to the bodies and have diameter at most one. Write the Hausdorff limit as K and the limiting disk intersection as L . It is immediate that $K \subset L$. Since the width-one limit K has interior, choose $z \in \text{int } K$. It lies strictly inside

every limiting disk. For $y \in L$, the points $(1 - \varepsilon)y + \varepsilon z$ lie strictly inside every limiting disk and therefore in all sufficiently late intersections. They belong to K in the limit. Letting $\varepsilon \downarrow 0$ proves $L \subset K$. This justifies convergence of the intersections instead of assuming it for arbitrary compact sets.

For density, choose $2m$ equally spaced support directions. Their unique contacts form an inner polygon I_m , and the lines form an outer polygon Q_m . Corner contact repetitions and zero outer sides are permitted. The paired-arc replacement described in the lecture is performed on each antipodal normal interval. For the interval from contacts a to b , draw the two unit circles about a, b and use their intersection toward the opposite contacts. The replacement's normal ranges pair the new corner with the opposite circular arc; the two endpoint pieces are paired with the old endpoint corners. It preserves the chosen contacts and supporting lines. Perform this on all intervals; repeated contacts contribute empty arcs and redundant centers are discarded. This gives a finite disk intersection of width one between I_m and Q_m .

Here is a quantitative squeezing argument from the contact triangles. In a triangle formed by two successive tangents and their contact chord, the vertex angle is $\pi - \pi/m$ and the chord is at most one. Its altitude to the chord is at most $\frac{1}{2} \tan(\pi/(2m))$ (the largest altitude for a fixed base and vertex angle occurs in the isosceles triangle). The perpendicular foot lies on the chord because this vertex angle is obtuse. Hence every outer vertex is within this distance of I_m , and convexity gives

$$d_H(I_m, Q_m) \leq \frac{1}{2} \tan(\pi/(2m)) \longrightarrow 0.$$

Both K and the constructed Reuleaux polygon are squeezed between these two polygons, so the same bound controls their Hausdorff distance. For width w , multiply the bound by w .

Blaschke motions and the classical three-arc reduction

To see a genuine two-vertex motion, let $x_0(t)$ move at unit speed on the circle about fixed x_k , in the direction lengthening the arc from x_{n-1} to x_0 . Adjust $x_{k+1}(t)$ by the other circle intersection. Taking the scalar product of the gradient with this tangent velocity yields

$$A'(t) = \sin \theta_0(t) \left(\tan \frac{\theta_{k+1}(t)}{2} - \tan \frac{\theta_0(t)}{2} \right).$$

The local geometry shows that θ_0 increases and θ_{k+1} decreases along this orientation. Once $\theta_0 \geq \theta_{k+1}$, the area decreases; equality is a local maximum along this motion. Relabel or reverse the appropriate motion to obtain a decreasing branch. Continue only while the Reuleaux combinatorics is valid. At its endpoint an arc collapses (or the triangle configuration is reached); redundant vertices are removed. Repeating reduces to the triangle. This is the differential version of the Blaschke deformation argument; the compactness plus stationarity proof above avoids having to implement that whole process.

The construction in Yaglom–Boltyanskii, Exercise 7-20, is a finite three-arc replacement. Denote a shortest arc by AB , its center by P , and the adjacent arcs at P by CP (center A) and PD (center B). Replace these three arcs by AQ (center C), QB (center D), and CD (center Q), all of radius one. The endpoint diameter conditions preserve constant width. The chord CD is longer than AB : the angles APC and BPD exceed 60° , while $APB < 60^\circ$ unless the shape is already a triangle. Therefore $CPD > 60^\circ$, so CD exceeds at least one of CP, PD , each at least AB by the shortest-arc choice.

The geometric comparison lemma needed here states that the curvilinear region removed near CPD has greater area than the region inserted near ABQ when $CD > AB$. Its proof is a separate diagram argument in the book, pp. 260–262: split both regions along PQ , pair the equal-radius pieces, and compare by inclusion when the two arc excesses have the same sign. For

opposite signs, slide the equal-radius pair; the increasing piece grows while the opposing piece shrinks, giving the strict difference comparison. Treat this comparison as a supplied geometric lemma if using this alternative in class; the course’s tangent-polygon and variational proofs do not depend on it. At least one replacement arc merges with an existing adjacent arc, so the number of arcs decreases. Iteration proves the inequality for Reuleaux polygons and density extends it. This is a useful picture of the mechanism, but do not present the unproved comparison lemma as an immediate consequence of the chord inequality.

Source and verification notes. The two elementary geometric comparisons follow Yaglom–Boltyanskii’s printed pp. 247–253 and 260–264. The derivative and multiplier perspectives follow Bogosel, [arXiv:2412.13808](https://arxiv.org/abs/2412.13808), Sections 3–4; we use squared diameter constraints only where their gradient agrees with the unit-distance version. The two new notebooks test analytic support identities, cut losses, regular-polygon areas, feasibility and the constrained derivative. Numerical tests illustrate these proofs; they do not establish global optimality.